

problem 970

Starting at zero, a kangaroo hops along the real number line in the positive direction. Each successive hop takes the kangaroo forward a uniformly random distance between 0 and 1. Let $H(n)$ be the expected number of hops needed for the kangaroo to pass n on the real line.

For example, $H(2) \approx 4.67077427047$. The first eight digits after the decimal point that are different from six are 70774270.

Similarly, $H(3) \approx 6.6665656395558899$. Here the first eight digits after the decimal point that are different from six are 55395558.

Find $H(10^6)$ and give as your answer the first eight digits after the decimal point that are different from six.

```
In [73]: def hops(n):
          d = 0
          number = 0
          while d < n:
              d += np.random.random(1)
              number += 1
          return(number)

          def H1(n):
              l = np.zeros(100000000)
              for i in len(l):
                  l[i] = hops(n)
              return(np.mean(l))
```

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In [74]: H1(2)
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IndexError                                Traceback (most recent call last)
Cell In[74], line 1
----> 1 H1(2)

Cell In[73], line 12, in H1(n)
     10 l = np.zeros(100000000)
     11 for i in len(l):
--> 12     l[i] = hops(n)
     13 return(np.mean(l))

IndexError: only integers, slices (`:`), ellipsis (`...`), numpy.newaxis (`None`) and integer or boolean arrays are valid indices
```

```
In [71]: def H2(n):
          avg = hops(n)
          for i in range(1,100000000):
```

```
    avg = ((i)*avg + hops(n))/(i+1)
    return(avg)
```

In []: H2(2)

In []: